

Personal Expense Tracker MCP

A local MCP server for adding, reviewing, updating, deleting, and summarizing personal expenses.

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3–5 MINUTES



The problem → a focused solution

Why we built the Personal Expense Tracker MCP

PROBLEM

- Expense records can be scattered across notes and messages.
- Finding a specific expense or calculating totals manually is slow.
- An AI client should not receive unlimited access to files, commands, or the internet.

SOLUTION

- Five focused MCP tools handle the core expense actions:
`add_expense` • `list_expenses` • `get_spending_summary`
`update_expense` • `delete_expense`
- Inputs are validated before the expense logic runs.
- Data stays in one fixed local file: `data/expenses.csv`.

Spending Summary — My Project Part

I focused on `get_spending_summary`.

- Connected the MCP tool to the real local expense data.
- Calculated monthly total amount, expense count, and totals grouped by category.
- Validated the tool input and verified the structured output.
- Tested the complete flow in MCP Inspector using August 2026 data.

Verified result: \$165 total • 3 expenses • category totals

TOOL INPUT

`get_spending_summary`

READ-ONLY

get a spending summary

month

A calendar month in YYYY-MM format

2026-08

Execute Tool

REAL RESULT

× Results

```
{
  "month": "2026-08",
  "totalAmount": 165,
  "expenseCount": 3,
  "categoryTotals": {
    "food": 45,
    "transport": 20,
    "bills": 100
  },
  "message": "spending summary created"
}
```

MCP Inspector proof: `get_spending_summary` executed successfully on local data.

Live demo: exactly two prompts

The live section shows one core write action and the spending-summary feature

5 TOOLS

add_expense

list_expenses

get_spending_summary

update_expense

delete_expense

PROMPT 1

Add a \$25 groceries expense for milk and bread on August 16, 2026.

PROMPT 2

Show my spending summary for August 2026.

MY FEATURE

Backup prompt

Summarize August 2026 spending and show the category with the highest total.

Tool: [get_spending_summary](#)

Next steps & demo takeaway

Future improvements after the complete project journey

01

Encrypted backup

Optional backup while keeping the local-first design.

02

Better insights

Clearer spending charts and summaries for users.

03

Safe CSV import

Preview and validate imported data before writing.

Takeaway: focused MCP tools + strict validation + safe local storage + bounded permissions.

READY FOR QUESTIONS